

State-Dependent Market (In)Efficiency in Cryptocurrency Markets

Sasan Barak^{*1}, Siavash Razmi², and Alireza Mousavi¹

¹*Department of Decision Analytics and Risk, Southampton Business School, University of Southampton, Southampton, UK*

²*Department of Computer Science, University of Innsbruck, Innsbruck, Austria*

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Abstract

Is the inefficiency in cryptocurrency markets constant or state-dependent? We test this using an adaptive framework that learns from novel on-chain and trend-based information. We find an adaptive approach not only yields superior risk-adjusted performance but also significantly reduces downside risk compared to static benchmarks. The source of this outperformance is economically intuitive; the model learns to prioritize behavioral trend signals during volatile regimes and fundamental on-chain metrics during stable periods. This validates these novel information sets as drivers of predictability, providing strong evidence for the Adaptive Market Hypothesis.

Keywords: cryptocurrency, market efficiency, adaptive market hypothesis, meta-learning, on-chain data, directional change, regime-dependent predictability

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^{*}Corresponding Author: s.barak@soton.ac.uk

1 Introduction

The emergence of cryptocurrencies has significantly challenged the fundamental assumptions of financial economics, particularly the Efficient Market Hypothesis (EMH) (Fama 1970). This new asset class, characterized by extreme volatility, innovative market microstructure, and unique digital transparency through public blockchains, provides a new space for examining price discovery and predictability.

The academic literature reveals a contentious debate over cryptocurrency market efficiency. Some studies demonstrate the variable and dynamic characteristics of inefficiency (Aslam et al. 2023; Meng and Khan 2024), while others indicate that the degree of efficiency is fundamentally connected to market conditions, including liquidity, volatility, and crisis periods (Bouteska et al. 2025), with the optimal investment strategy itself being regime-dependent (Salis and Maciel 2023). However, competing evidence documents increasing efficiency through rapid arbitrage mechanisms (Makarov and Schoar 2020) and more integrated price discovery (Brauneis and Mestel 2018), particularly in major cryptocurrency pairs, challenging the persistent inefficiency narrative.

This paper argues that these conflicts arise because predictability in cryptocurrency markets is not static but highly regime-dependent. Our perspective is aligned with the Adaptive Market Hypothesis (AMH) (Lo 2004), which posits that market efficiency evolves as participants adapt and learn. We move beyond traditional diagnostics of market inefficiency to test a direct, economically motivated hypothesis: opportunities for abnormal returns are temporary, emerging and dissipating with market regimes, and driven by information sets unique to digital assets. Our central contribution is to identify, model, and empirically validate these sources through a meta-learning framework that attributes predictability to specific, interpretable features.

Specifically, we investigate two novel information sources: (1) fundamental network activity, measured using on-chain blockchain data such as transaction volumes and hash rates; and (2) complex behavioral patterns, captured through asymmetries in market trends and directional movements. These information sets, we argue, provide unique insights into the underlying dynamics of cryptocurrency markets that are not captured by traditional financial metrics.

To test this hypothesis, we construct a sophisticated empirical framework combining three key methodological innovations. First, acknowledging data integrity challenges inherent in 24/7 markets (Vidal-Tomás 2021), we employ the Directional Change (DC) methodology, which defines price movements in intrinsic, event-based time triggered by moves exceeding threshold θ , rather than fixed calendar intervals (Tsang et al. 2017). This approach enables more economically meaningful identification of trends and reversals, particularly well-suited to cryptocurrency markets' high-volatility environment.

Second, we ensure market-wide generalizability through portfolio construction using symmetric uncertainty (SU) clustering, which captures non-linear asset dependencies more effectively than traditional correlation-based methods. This technique allows us to construct diversified portfolios that represent broader market phenomena rather than asset-specific characteristics.

Third, our meta-learning framework employs supervised learning to dynamically select optimal trading parameters by integrating four economically interpretable information sets: his-

torical strategy performance, on-chain blockchain metrics, DC-derived trend indicators, and traditional market statistics. The framework creates a direct comparison between adaptive and static implementations of the same underlying trading logic, enabling rigorous quantification of the economic value of modeling state-dependency.

Our empirical results, based on high-frequency data for 50 major cryptocurrencies and evaluated through rigorous out-of-sample testing, strongly support our hypothesis. The adaptive framework achieves a Sharpe ratio more than double that of the best-performing static benchmark, a result that is both statistically and economically significant that strongly supports the AMH by quantifying regime-dependent alpha.

In doing so, this paper makes the following contributions:

- We provide robust, out-of-sample evidence supporting the AMH in cryptocurrency markets, demonstrating that predictability is time-varying and regime-dependent.
- We introduce and validate two novel information sets fundamental on-chain data and DC-derived trend asymmetries as key economic drivers of this state-dependent inefficiency.
- We propose a novel framework for applying the DC methodology to a multi-asset portfolio using SU clustering, bridging a gap between event-based analysis and practical portfolio management.
- We develop a return-weighted loss function for our meta-learning framework, introducing a methodological refinement that better aligns the model’s statistical objective with an investor’s economic goals.
- We conduct a detailed attribution analysis using SHAP, opening the "black box" of the adaptive model to provide an economic interpretation of its learned, state-dependent decision rules.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature on cryptocurrency market efficiency and regime-dependent trading strategies. Section 3 details our data sources and empirical methodology, including the meta-learning framework and portfolio construction techniques. Section 4 presents our main empirical results and extensive robustness checks. Section 5 discusses the implications of our findings for market efficiency theory and practical trading applications. Section 6 concludes.

2 Literature Review

The rapid rise of cryptocurrencies as a distinct asset class has sparked a sustained debate in quantitative finance on market efficiency, behavioral dynamics, and portfolio design. This review places our study in that debate, with a focus on the tension between the Efficient Market Hypothesis (EMH) and the AMH, the role of regime-dependent predictability, and the promise of event-based sampling and adaptive methods. In line with our introduction, we emphasize that predictability in digital assets is time-varying and state-dependent, that on-chain transparency and trend asymmetries supply distinct information, and that portfolio evidence on regime dynamics is most convincing when tested out-of-sample.

2.1 Crypto Market Efficiency: EMH vs. AMH

At the core of cryptocurrency research lies the question of market efficiency, rooted in Fama (1970). Early studies challenged weak-form efficiency, documenting autocorrelation and long memory (Urquhart 2016; Nadarajah and Chu 2017), while other work linked deviations from efficiency to liquidity and volatility (Brauneis and Mestel 2018; Corbet et al. 2019; Abou Tanos and Badr 2024). This has led to a contentious debate. Some evidence points to faster arbitrage and rising efficiency (Makarov and Schoar 2020; Brauneis and Mestel 2018). However, other research provides a stark counter-narrative, documenting substantial, unexploited arbitrage opportunities and concluding that bitcoin market inefficiency actually increased over time (Krückeberg and Scholz 2020). This conflict has led to a growing consensus that market efficiency is not a static condition but a dynamic, time-varying process a view aligned with the AMH of Lo (2004). The AMH suggests that efficiency fluctuates as market conditions change and as investors adapt to new information and external shocks (Li, Manahov, and Ashton 2025; Vukovi et al. 2024). This perspective is now strongly supported by a large body of empirical work demonstrating that cryptocurrency markets alternate between periods of efficiency and inefficiency (Mokni et al. 2025). Major global events serve as natural experiments for the AMH; the COVID-19 pandemic and the Russia-Ukraine War, for instance, have been shown to significantly impact market structures, often intensifying multifractality and leading to periods of greater inefficiency and herd behavior across financial markets (Mensi et al. 2025; Ahmed Memon et al. 2024; Takaishi 2025). Similarly, internal market shocks like major cryptocurrency heists are also shown to significantly disrupt market efficiency and trigger maladaptive investor behavior (Li, Manahov, and Ashton 2025).

2.2 Regime-Dependent Dynamics: Multifractality and Its Economic Drivers

The time-varying nature of cryptocurrency market efficiency is most commonly quantified using multifractal detrended fluctuation analysis (MFDFA). This statistical tool is well-suited to the volatile and non-stationary nature of crypto assets and has been used extensively to provide robust evidence of multifractal properties a key signature of market inefficiency and long-range dependence. Numerous recent studies employ MFDFA and its asymmetric variants to provide robust evidence that multifractality is a core feature of crypto markets and its degree varies significantly over time, often in response to crises (Almeida, Perlin, and Müller 2025; Mensi et al. 2025; Ahmed Memon et al. 2024; Takaishi 2025). Salis and Maciel (2023) show that this measured inefficiency has a direct, regime-dependent impact on portfolio outcomes, confirming its economic relevance.

While MFDFA is excellent for describing the statistical footprint of inefficiency, a growing body of research now focuses on identifying its underlying economic drivers. Liquidity and volatility are consistently highlighted as primary determinants. For instance, Al-Yahyaee et al. (2020) show that both liquidity and volatility are significant drivers of inefficiency in cryptocurrency markets, a finding consistent with evidence from traditional equity sectors where higher volatility and lower trading volume also indicate stronger predictability (Park and Newaz 2023). More recently, Mokni et al. (2025) find that while higher liquidity improves efficiency across all market states, the positive effect of money flow is significant only when markets are

already efficient. Using a price delay metric, Abou Tanos and Badr (2024) confirm that efficiency is stronger in more liquid and less volatile cryptocurrencies, especially during crises. The classic price-volume relationship is also critical; studies suggest trading volume acts not as a direct proxy for information flow but rather as an indicator of its *quality* and dispersion, making it a significant predictor for volatility but not necessarily returns (Patra and Gupta 2025; Sahoo and Sethi 2024). Furthermore, the regulatory environment plays a crucial role, with evidence showing that regulatory compliance significantly enhances market efficiency and reduces investor risk (Nimalendran et al. 2025).

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2.3 Directional Change for Event-Based Analysis

The widespread evidence of asymmetric multifractality in cryptocurrency markets, where downside and upside trends exhibit different scaling properties (Mensi et al. 2025), underscores the limitations of standard fixed-interval analysis. A methodology like DC, which defines trends based on intrinsic, event-driven price movements exceeding a certain threshold, is therefore uniquely suited to capture these economically meaningful asymmetries (Tsang et al. 2017; Tsang, Tao, and Ma 2015). This approach aligns well with how trading decisions are made in practice and is particularly well-suited to the high-volatility, non-stationary environment of cryptocurrency markets. While portfolio-level DC applications remain scarce, its ability to extract trend-based signals that align with the documented fractal nature of crypto markets makes it a promising tool (Salis and Maciel 2023).

2.4 Meta-Learning for Market Adaptation

Our meta-learning approach aligns with a recent turn in the literature toward using advanced deep learning and machine learning frameworks to move beyond static tests of efficiency and formally model the AMH. For example, Vukovi et al. (2024) use Long Short-Term Memory (LSTM) models to identify temporary "pockets in time" of market inefficiency that challenge the EMH. Polyzos, Rubbaniy, and Mazur (2024) train classifiers on social media data to build a dynamic, information-based index of market efficiency, directly testing the fundamental definition of efficiency as related to public information. While a separate branch of literature uses deep learning to detect asset price bubbles from option prices (Biagini et al. 2025), the common thread is the use of adaptive algorithms to model and detect state-dependent market behavior. Our meta-learning framework extends this by not only detecting regimes but also creating a direct, out-of-sample test of the economic value of adapting to them, in line with the established practice of identifying economic regimes to improve performance in traditional asset management (Mulvey and Liu 2016).

2.5 Gaps and Contributions

The literature robustly shows that cryptocurrency efficiency is time-varying, as predicted by the AMH. This dynamic inefficiency is often described using multifractal analysis and is driven by a complex interplay of liquidity, regulation, information flow, and external shocks. Yet three key gaps remain. First, event-based sampling that captures trend asymmetries in real time is underused in portfolio settings. Second, while research has explored novel information sources like corporate political connections (Kim et al. 2018), social media sentiment (Polyzos, Rubbaniy, and Mazur 2024), ESG factors (Fereydooni, Barak, and Sajadi 2024), and the impact of shocks like cryptocurrency heists (Li, Manahov, and Ashton 2025), the fundamental, verifiable information from on-chain data remains underutilized in dynamic portfolio frameworks. Third, and most critically, the majority of the literature remains descriptive. Few studies have moved from characterizing inefficiency to building an integrated, adaptive framework that models its drivers and formally tests its economic exploitability in a rigorous, out-of-sample portfolio context. For instance, Mokni et al. (2025) document long memory and liquidity-driven inefficiency, particularly in efficient regimes, motivating our use of on-chain flow metrics to capture sentiment shifts. Similarly, Mensi et al. (2025) highlight asymmetric multifractality during crises, justifying DC indicators as proxies for behavioral trend asymmetries.

Our study addresses these gaps by (i) aligning with the AMH and testing a direct state-dependent hypothesis in portfolios, (ii) integrating on-chain and DC-derived features to detect and exploit regimes, (iii) ensuring market-wide generalizability via SU clustering for portfolio construction, and (iv) using a supervised meta-learning layer to adapt parameters over time. In doing so, we reconcile competing findings on efficiency and deliver robust out-of-sample evidence on the economic value of modeling state-dependency in cryptocurrency markets.

3 Data and Empirical Framework

Drawing from the AMH, we hypothesize that cryptocurrency market inefficiency varies across regimes, driven by novel signals such as on-chain data and event-driven trend asymmetries. Our framework rigorously tests this hypothesis out-of-sample by measuring the risk-adjusted economic value of adaptive strategies relative to static benchmarks, thereby extending studies on cryptocurrency market efficiency (Urquhart 2016). To this end, we propose a multi-asset portfolio framework that confronts two key methodological challenges: first, building a portfolio that robustly captures non-linear co-movements among cryptocurrency returns; and second, ensuring methodological alignment between our multi-asset portfolio and the DC framework. This section presents our innovative solutions.

3.1 Data and Sample Construction

3.1 High-Frequency Cryptocurrency Data

Our empirical analysis relies on a high-frequency dataset constructed from multiple sources to capture the regime-dependent volatility and non-stationarity of the cryptocurrency market, consistent with evidence of its frequent regime shifts and microstructure noise (Dimpfl and Peter

2021; Fang et al. 2024). The sample period spans January 2021 to January 2024, encompassing bull and bear cycles that reflect the market’s evolving dynamics. The primary price data consists of minute-level futures market data sourced from the Binance exchange. Our trading universe is comprised of the top 50 cryptocurrencies by market capitalization representative of the sample period (January 2021 to January 2024), as reported by [CoinMarketCap 2024](#). We use futures data as it provides a centralized and highly liquid venue, which avoids the data fragmentation and reliability issues common across disparate spot exchanges Vidal-Tomás 2021. This focus on a consistent universe of liquid assets ensures the stability of our subsequent clustering and portfolio selection process.

3.1 On-Chain Data

A central innovation of our study is the integration of on-chain blockchain data, which we argue serves as a proxy for fundamental network activity and investor behavior, offering an information set distinct from traditional price and volume data. We collected approximately 300 on-chain metrics for the Bitcoin (BTC) and Ethereum (ETH) networks from [Glassnode 2024](#).

These networks were selected as they are the two most dominant crypto-assets by market capitalization and network value, ensuring our analysis is grounded in the most economically significant and data-rich digital assets.

To ensure a concise and effective model, we employed a feature selection process based on the importance scores generated by an initial meta-learning model, reducing the initial set to the 43 most informative metrics. To prevent look-ahead bias, this feature selection process was conducted within the walk-forward validation loop, using only the training data available at each step. These final features, detailed in Appendix A, fall into seven distinct economic categories:

- **Supply:** Metrics related to the holding behavior of investors, such as the proportion of coin supply that has been inactive for various periods.
- **Exchange:** Data capturing the flow of assets into and out of centralized exchanges, which can indicate shifts in investor sentiment from holding to selling.
- **Transactions:** Measures of network usage, including the number of sending addresses and average transaction fees, reflecting demand for block space.
- **Mining:** Indicators of network security and production costs, such as the aggregate hash rate and miner revenue.
- **Valuation:** Ratios that compare market price to on-chain valuation models, like the Market Value to Realized Value (MVRV) Z-score, which can identify periods of market over- or undervaluation.
- **Spent Outputs & Activity:** Metrics tracking the volume of coins being moved after long periods of dormancy, signaling actions by long-term holders.

- **Price Indicators:** On-chain derivatives of price, such as regional price changes during specific trading hours and realized volatility.

The economic rationale for the predictive power of these on-chain metrics is grounded in their ability to provide a transparent, near-real-time view into the fundamental drivers of network value and investor behavior information not fully captured by traditional market data, consistent with the AMH. For instance, Supply metrics reflecting long-term holder behavior serve as a proxy for the conviction of sophisticated or core investors (e.g., heterogeneous beliefs between miners and traders); a rising proportion of dormant supply may signal a positive long-term outlook amid lower perceived risk. Exchange flows can be interpreted as a direct measure of market sentiment, where large inflows often precede sell-offs, indicating a "fear signal" from retail investors shifting to a speculative stance. Similarly, Mining metrics like the hash rate are a direct measure of network security and capital investment, where a rising hash rate can lower perceived long-term risk and justify momentum strategies. Finally, Valuation metrics like the MVRV Z-score explicitly model periods of speculative excess by comparing an asset's market price to its on-chain cost basis, providing a fundamental anchor that helps identify market bubbles and bottoms driven by sentiment among long-term holders. By integrating these features, we are not merely using alternative data; we are testing whether the market efficiently incorporates information about the underlying health, security, and sentiment of the digital economy itself, as posited by AMH.

3.2 An Event-Based Approach to Time: The Directional Change Methodology

3.2 Economic Motivation for Event-Based Time

Traditional financial econometrics relies on data sampled at fixed calendar intervals, an assumption that is systematically violated in high-frequency markets where "stylized facts" like volatility clustering are pronounced Ghosh et al. 2023. To overcome this, we adopt the DC framework, an innovative approach that samples data in an intrinsic, event-based time Tsang et al. 2017; Tsang, Tao, and Ma 2015. The DC methodology defines a new "event" only when the price moves by a pre-determined significant magnitude, θ . This aligns the analysis with the economic reality of market participants and makes the sampling frequency endogenous to market volatility, providing an adaptive and economically coherent timescale. Furthermore, the DC framework's focus on trend asymmetries provides a powerful lens through which to examine market behavior. The documented asymmetric multifractality in cryptocurrency markets (Mensi et al. 2025; Ahmed Memon et al. 2024) suggests that investor reactions to upward and downward trends are not symmetric. This observation aligns with foundational concepts in behavioral finance, such as Prospect Theory (Kahneman and Tversky 2013), which posits that investors are more sensitive to losses than to equivalent gains. Such biases can create predictable over- and under-reactions; for example, recent work shows that investors tend to overreact to salient, extreme short-term price movements, leading to significant return predictability (Ko, Wang, and Yang 2025). In this context, our DC-derived indicators that measure the relative duration or magnitude of up-trends versus down-trends are designed to capture these types of cognitive biases as proxies for time-varying behavioral phenomena, leading to temporary,

exploitable inefficiencies under the AMH.

3.2 Formal Definition of Directional Change

The DC framework partitions a price series into a discrete sequence of alternating upward and downward trends, marked by DC events (Aloud et al. 2012). A new trend is confirmed at a Directional Change Confirmation (DCC) point at the moment the price reverses by a magnitude greater than the threshold, θ , from its last recorded extreme.

Formally, if the market is in a downward trend, a new upward trend is confirmed when the current price, P_c , breaches the threshold relative to the last low point, P_{low} :

$$P_c \geq P_{low} \times (1 + \theta) \quad (1)$$

Upon confirmation, the historical price P_{low} is retrospectively designated the Extreme Point (EXT) of the preceding downward trend. Conversely, if the market is in an upward trend, a new downward trend is confirmed when:

$$P_c \leq P_{high} \times (1 - \theta) \quad (2)$$

where P_{high} is the highest price recorded during the upward trend. The procedural logic for identifying these events is formally detailed in Algorithm 1, located in Appendix B.

Any price movement that continues in the direction of the new trend following a DCC point is defined as an Overshoot (OS) event. These core components are illustrated in Figure 1. For definitional clarity, all DC-related notations used throughout this paper are summarized in Table 1.



Figure 1: An illustration of the DC methodology. Point (a) is an extreme point (EXT), identified retrospectively once the Directional Change Confirmation (DCC) point at (b) occurs. The period between (a) and (b) constitutes the DC interval, while the period from (b) to the next extreme point at (c) is the Overshoot (OS) interval.

Our empirical framework extends this base methodology by employing a dual DC summarization technique, which simultaneously tracks trends using two distinct thresholds: a "small

Table 1: Overview of notations used in this study

Name/Description	Notation
Threshold for summarizing price movements	θ
Extreme point based on θ summary	EXT
Upward DC event based on θ threshold	$DC_{\theta}^{\uparrow}$
Downward DC event based on θ threshold	$DC_{\theta}^{\downarrow}$
OS event	OS
Current Price	P_c
Price at a DCC point based on θ^s	P_{DCC}^s
Price at extreme point based on θ^s	P_{EXT}^s
Previous high price in a downtrend	P_{high}
Previous low price in an uptrend	P_{low}
Timestamp of the DCC point based on θ^s	t_{DCC}^s
Timestamp of the extreme point based on θ^s	t_{EXT}^s
Timestamp of the previous high price during a downtrend	t_{high}
Timestamp of the previous low price during an uptrend	t_{low}
Target label for sample i in DC Prediction	y_i
Predicted label for sample i from our DC prediction model	$\hat{y}_i$

theta" (θ^s) and a "big theta" (θ^b), where $\theta^b > \theta^s$ (Bakhach, Tsang, and Jalalian 2016). This approach, visualized in Figure 2, allows us to create a more sophisticated prediction task for our primary trading model.



Figure 2: A dual DC summary using thresholds θ^s (green) and θ^b (red). The target variable for a given θ^s event, such as at DCC(a), is 1 because its corresponding extreme point EXT(a) is also an extreme point for the larger θ^b trend. The target for DCC(b) is 0 because its corresponding extreme point EXT(b) is not an extreme point for the θ^b trend.

The model's objective is to forecast whether a smaller trend, confirmed by θ^s , possesses enough momentum to extend into a more significant trend that would satisfy the larger threshold, θ^b . Our formulation of this binary prediction task (y_i) represents a key innovation. We extend the descriptive dual-threshold approach of Bakhach, Tsang, and Jalalian (2016) by transforming it into a formal test of trend persistence.

In our framework, a positive signal ($y_i = 1$) can be interpreted as evidence of strong market momentum, whereas a negative signal ($y_i = 0$) suggests an impending reversal, thereby connecting our event-based analysis to the broader literature on momentum and reversal anomalies

under the AMH. This is specified as a binary target variable, where the outcome is 1 if an extreme point under the θ^s regime is also an extreme point under the θ^b regime, and 0 otherwise:

$$y_i = \begin{cases} 1 & \text{if } P_{\text{EXT},i}^s \in \{P_{\text{EXT}}^b\} \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

where $P_{\text{EXT},i}^s$ is the extreme point for the i -th DCC point in the θ^s series, and $\{P_{\text{EXT}}^b\}$ is the set of all extreme points in the θ^b series.

3.3 Portfolio Construction and Market Representation

A central objective of this study is to evaluate the economic value of state-dependency in a robust and generalizable manner across the broader cryptocurrency market. While a single-asset analysis can be illustrative, it is vulnerable to asset-specific noise and may fail to capture systematic market phenomena. To address this, we develop a multi-asset portfolio framework. However, this approach introduces two key methodological challenges: first, constructing a portfolio that accurately captures the complex, non-linear co-movement structure of cryptocurrency returns; and second, ensuring methodological alignment between our multi-asset portfolio and the DC framework, which is inherently univariate. This section outlines our solutions to these challenges.

3.3 Asset Selection via Non-Parametric Clustering

The traditional approach to portfolio construction, rooted in Modern Portfolio Theory (Markowitz 1952), relies on covariance matrices derived from Pearson correlations. However, this method assumes linearity and normality conditions that cryptocurrency returns systematically violate (Corbet et al. 2019). To overcome these limitations, we adopt a non-parametric measure from information theory: SU (Witten et al. 2005; Liu and Garrett 2023). Unlike correlation, SU is robust to non-linear dependencies and requires no assumptions about the underlying data distribution. It quantifies the shared information between two variables using entropy, $H(X)$, and mutual information, $MI(A, B)$:

$$H(X) = - \sum_{x \in X} p(x) \log p(x) \quad (4)$$

where $p(x)$ represents the probability of each state x in a discrete random variable X . Mutual Information is then defined as:

$$MI(A, B) = H(A) + H(B) - H(A, B) \quad (5)$$

where $H(A)$ and $H(B)$ are the entropies of variables A and B , respectively, and $H(A, B)$ is their joint entropy. The SU metric normalizes the mutual information by the sum of the individual entropies, providing a stable, bounded measure of association from 0 (statistical independence) to 1 (perfect association) :

$$SU(A, B) = \frac{2MI(A, B)}{H(A) + H(B)} \quad (6)$$

Our portfolio construction is a dynamic process, re-executed monthly to adapt to the market’s evolving structure. Each month, we compute a pairwise SU matrix using daily returns from the preceding month for all 50 assets in our universe. This matrix serves as input for a hierarchical clustering algorithm, which is well-suited for identifying the nested co-movement structures common in financial assets. Figure 3 illustrates SU’s superior ability to reveal these underlying patterns compared to Pearson correlation.

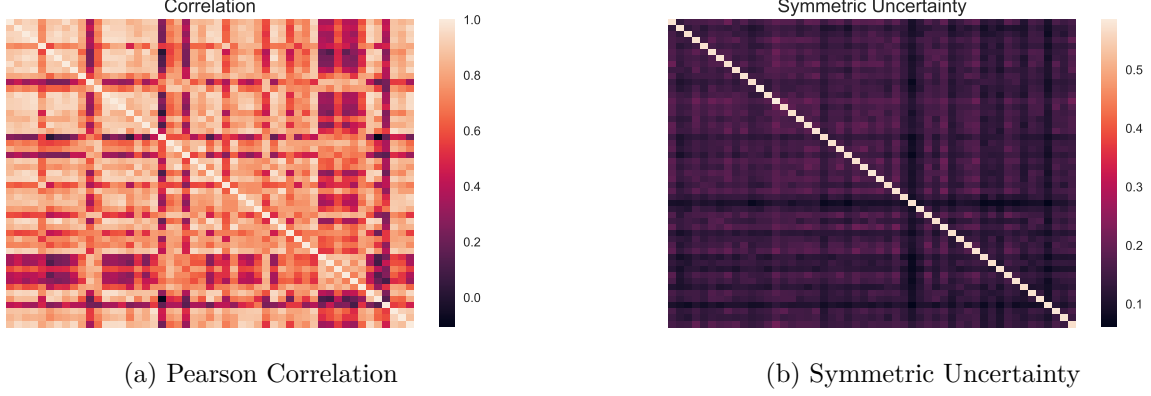


Figure 3: Comparison of asset relationship heatmaps using (a) Pearson Correlation and (b) Symmetric Uncertainty on daily returns, illustrating the superior clustering capability of the SU metric.

The optimal number of clusters is determined empirically using the silhouette method, which assigns a score s_n to each sample to measure cluster cohesion (Barak and Mokfi 2019):

$$s_n = \frac{b_n - a_n}{\max(a_n, b_n)} \quad (7)$$

where a_n is the average SU distance between sample n and other samples within the same cluster, and b_n is the average distance between sample n and objects in the nearest cluster. From each resulting cluster, we select the single most central asset as its representative. Capital is then allocated via an equally weighted ($1/N$) scheme, where the weight for each of the N_{assets} is given by:

$$W_i = \frac{1}{N_{\text{assets}}}, \quad \forall i \in \{1, 2, \dots, N_{\text{assets}}\} \quad (8)$$

This approach delivers a robust and neutral allocation while mitigating estimation risk in optimized covariance-based weightsa challenge exacerbated in high-dimensional, non-stationary environments such as cryptocurrency markets.

3.3 A Novel Aggregation Technique for Portfolio-Level Analysis

A key methodological contribution of this paper is a technique to apply the DC framework, natively designed for a single time series, to our multi-asset portfolio. This is achieved through a two-step price aggregation process. First, to standardize assets with vastly different price levels, the price of each portfolio constituent i , $P_i(t)$, is normalized by its price at the inception

of the analysis period, $P_i(0)$:

$$\tilde{P}_i(t) = \frac{P_i(t)}{P_i(0)} \quad (9)$$

This creates a set of unitized price series, each representing the cumulative return path of its respective asset. Second, these normalized price series are summed to create a single, aggregated portfolio price series, $P_T(t)$:

$$P_T(t) = \sum_{i=1}^{N_{\text{assets}}} \tilde{P}_i(t) \quad (10)$$

The resulting series, $P_T(t)$, has a clear economic interpretation: it represents the value evolution of a portfolio constructed by investing one unit of capital equally across all selected assets at $t = 0$ and tracking its performance. The DC methodology is then applied directly to this unified price series to identify economically significant trends, reversals, and overshoots at the aggregate portfolio level, thereby bridging the gap between our multi-asset universe and our event-based, univariate analytical framework.

3.4 A Meta-Learning Framework for Modeling State-Dependency

The central thesis of this paper engages with the debate between the static Efficient Market Hypothesis (EMH) and the dynamic Adaptive Market Hypothesis (AMH). The AMH posits that market efficiency varies over time, creating transient opportunities for abnormal returns that emerge and dissipate with market regimes. To formally test the economic implications of the AMH, we develop a meta-learning framework that moves beyond a static model by dynamically adapting its strategy in response to these changing conditions. This approach provides a direct, out-of-sample quantification of the economic value derived from modeling state-dependency.

3.4 The Economic Hypothesis: Adaptive vs. Static Strategies

We formalize the core tension between the EMH and the AMH into a directly testable economic hypothesis. Our null hypothesis (H_0) is motivated by weak-form efficiency and posits that no adaptive strategy can consistently outperform the best-performing static strategy on a risk-adjusted basis. Our alternative hypothesis (H_1), motivated by the AMH, is that time-varying market conditions create predictable, regime-dependent inefficiencies that an adaptive strategy can exploit. We specify these as:

H_0 (EMH): *The Sharpe ratio of the adaptive meta-learning strategy is not statistically distinguishable from the Sharpe ratio of the best-performing static θ strategy.*

H_1 (AMH): *The Sharpe ratio of the adaptive meta-learning strategy is statistically and economically greater than the Sharpe ratio of the best-performing static θ strategy.*

Our meta-learning framework is the empirical mechanism designed to test H_1 against H_0 in a rigorous, out-of-sample setting.

3.4 Feature Engineering for State-Dependency

To formally test our hypothesis that market predictability is state-dependent, we engineer four distinct sets of features to serve as inputs for our meta-learning framework. Each feature set is designed to capture a different facet of the market’s economic state. The complete mathematical definitions for these features are provided in Appendix J.

1. **Strategy Meta-Information:** Metrics reflecting the recent historical performance of the underlying trading strategies, such as win rates, number of trades, and average returns.
2. **On-Chain Information:** The curated set of 43 fundamental blockchain metrics described above, capturing the underlying health and activity of the networks.
3. **DC-Derived Indicators:** A comprehensive set of indicators derived from the DC methodology, designed to quantify trend strength, duration, and asymmetries in market movements.
4. **Market Regime Information:** Traditional statistical measures of market behavior, including the mean, standard deviation, and skewness of prices, as well as standard technical indicators.

3.4 The Meta-Learning Architecture

Our framework is constructed as a two-level learning system, a concept broadly known as meta-learning or "learning to learn" (Vanschoren 2018). The first level consists of a set of distinct “base” trading strategies. Each base strategy is a full implementation of the portfolio trading logic described previously but operates with a different, fixed DC threshold, θ^s , drawn from a discrete set $\Theta = \{\theta_1, \theta_2, \dots, \theta_m\}$. These base strategies represent competing, static models of market behavior. The specific features and trading logic used in these base models are formally defined in Appendix C.

The second level is the meta-learning model, which acts as a dynamic strategy selection algorithm. Its objective is to predict, at the end of each day d , which of the base strategies will yield the highest return on the subsequent day, $d + 1$. This is framed as a classification task where the ground truth label, θ_*^s , is the parameter that ex-post generated the highest return:

$$\theta_*^s = \arg \max_{\theta_i \in \Theta} \text{Return}(\theta_i, d) \quad (11)$$

The meta-learning model, f_{meta} , maps the feature set $\mathbf{m}_d$ —representing the observed market state at day d —to a predicted optimal parameter for the next day, $\hat{\theta}_{d+1}^s$:

$$\hat{\theta}_{d+1}^s = f_{\text{meta}}(\mathbf{m}_d) \quad (12)$$

The model is trained on a history of feature-label pairs, $(\mathbf{m}_d, \theta_{d+1}^s)$, learning the relationship between observable market characteristics and subsequent strategy performance. The information set, $\mathbf{m}_d$, comprises the four distinct categories of features detailed in Section 3.1: (1) Strategy Meta-Information, (2) On-Chain Information, (3) DC-Derived Indicators, and (4) Market Regime Information.

3.4 A Novel Weighted Loss Function for Economic Significance

A fundamental challenge in applying standard classification algorithms to financial tasks is the mismatch between their statistical objective functions and an investor’s economic objectives, a dynamic empirically demonstrated in Appendix D. A typical classifier is trained to maximize accuracy, treating all misclassifications as equally costly (Bishop 2006). In trading, however, the economic consequence of a decision is paramount. Mis-selecting a parameter on a day when all strategies perform similarly is economically trivial; making the wrong choice on a day with high performance dispersion can be catastrophic.

To align the model’s training process with economic reality, we introduce a novel, return-based weighting function for the model’s loss function. This approach ensures that the model pays significantly more attention to training samples where the choice of parameter has the greatest financial impact. The weight for each daily sample, w_d , is an exponential function of the difference between the maximum and minimum returns observed across the set of base strategies for that day:

$$w_d = \exp(\alpha \cdot (r_{\max,d} - r_{\min,d})) \quad (13)$$

Here, $r_{\max,d}$ and $r_{\min,d}$ are the highest and lowest returns on day d , and α is a scaling parameter. To mitigate the influence of extreme outliers, daily returns are clipped at three standard deviations from their mean before calculating the weights. This weighting scheme transforms the standard machine learning objective function into an economically motivated one. The updated loss function, $\mathcal{L}$, is the weighted sum of the individual sample losses, $\ell(y_i, \hat{y}_i)$:

$$\mathcal{L} = \sum_{i=1}^n w_i \cdot \ell(y_i, \hat{y}_i) \quad (14)$$

By incorporating this function, we train the meta-learning model to be most sensitive to the instances that matter most for portfolio performance, thereby creating a more robust and economically-tuned adaptive framework.

3.5 Empirical Design and Out-of-Sample Validation

3.5 Walk-Forward Validation for Robustness

Standard statistical cross-validation techniques, such as k-fold, are invalid for financial time series as they fail to preserve the temporal ordering of data, introducing look-ahead bias and leading to spuriously inflated performance estimates (De Prado 2018). We therefore employ a walk-forward validation methodology to simulate a realistic, out-of-sample trading environment where all decisions are based solely on information that would have been available historically.

We utilize two distinct walk-forward schemes, each tailored to the specific objective of the model being trained:

1. **Rolling-Origin for the Primary Strategy:** The base DC prediction models are trained on a rolling window. This prioritizes recency, ensuring the model is continually recalibrated on the most recent market data, which is essential for the short-term DC event forecasting task.

2. **Fixed-Origin for the Meta-Learning Model:** The meta-learning model is trained on an expanding window. This prioritizes a long-term memory, which is crucial for the meta-learner’s task of identifying recurring, large-scale market regimes that may have appeared at any point in the sample history.

These two validation methodologies are illustrated in Figure 4.

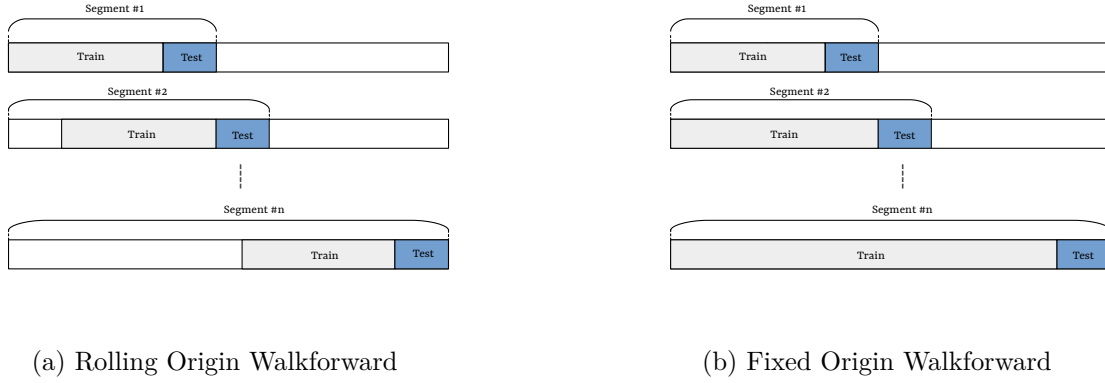


Figure 4: Walk-forward validation schemes. Panel (a) shows the Rolling Origin approach for the primary strategy. Panel (b) shows the Fixed Origin approach for the meta-learning model.

3.5 Performance Evaluation Metrics

The economic performance of both the static and adaptive strategies is evaluated using a comprehensive suite of metrics designed to provide a multi-faceted view of profitability, risk-adjusted returns, and downside risk. Our primary evaluation metrics include: Rate of Return (RR), Profit Factor (PF), Maximum Drawdown (MDD), Longest Drawdown Duration (LDD), the Sharpe Ratio (SRPR), and the Sortino Ratio (SORR). The formal definitions of these metrics are provided in Appendix E for reference.

3.5 Attribution Analysis with SHAP

A common and valid critique of machine learning models in finance is their “black box” nature. To ensure our results are not only statistically robust but also economically interpretable, we move beyond performance reporting and conduct a rigorous attribution analysis. We employ SHAP (Shapley Additive Explanations), a framework rooted in cooperative game theory that attributes the value of a model’s prediction among its input features (Lundberg and Lee 2017).

This analysis is a critical component of our paper, as it allows us to understand the economic drivers behind the meta-learning model’s decisions. By examining SHAP values, we can identify which information sets—on-chain data, DC-derived trend asymmetries, or traditional market statistics—are most influential in determining the optimal strategy parameter during different market regimes. This provides a direct empirical link between the sources of market inefficiency we hypothesize and the adaptive strategy’s ability to generate excess returns, transforming the model’s outputs into actionable economic insights.

4 Empirical Results

4.1 Experimental Design and Static Model Performance

We first establish a benchmark by implementing the primary trading strategy with a single, static θ^s parameter, optimized over a dedicated in-sample period (January 2021 to January 2022). The out-of-sample performance of this optimized static model, which uses $\theta^s = 0.026$, is presented in Table 2.

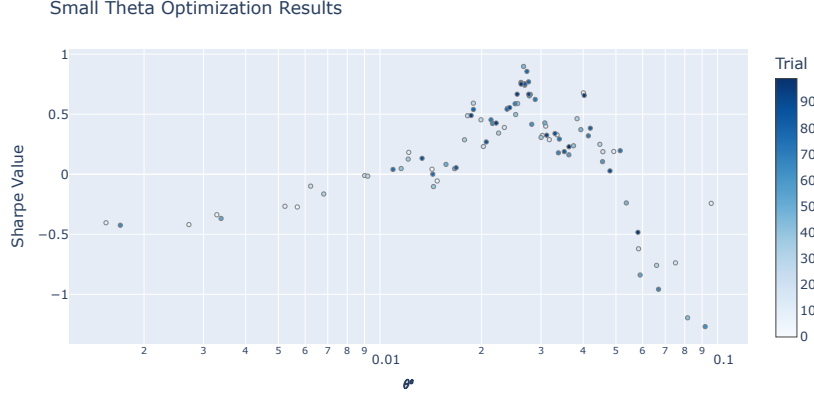


Figure 5: Optimization of Static θ^s Parameter

Notes: This figure displays the Sharpe ratio achieved during the in-sample optimization period (Jan 2021—Jan 2022) for different values of the static θ^s parameter. The optimal value of 0.026 is selected for the static benchmark strategy.

The results in Table 2 reveal a key challenge. While our SU-based portfolio construction adds value relative to standard benchmarks like Equally Weighted (EW) and Hierarchical Risk Parity (HRP), the static portfolio strategy is outperformed on both an absolute and risk-adjusted basis by a single-asset implementation on Bitcoin (Sharpe ratio of 0.46 vs. 0.63). This finding highlights that a single static parameter fails to generalize across diverse assets and evolving market conditions, providing a strong economic motivation for the adaptive, state-dependent framework we introduce next.

Table 2: Out-of-Sample Performance of the Optimized Static Strategy

Metric	Portfolio ($\theta^s = 0.026$)	Bitcoin	EW	HRP	HERC
Rate of Return (%)	31.25	75.96	−19.05	−17.79	−17.76
Max Drawdown (%)	−56.31	−37.90	−66.20	−65.97	−65.95
Profit Factor	1.09	1.18	1.01	1.02	1.02
Sharpe Ratio	0.46	0.63	0.07	0.08	0.08
Sortino Ratio	0.63	0.94	0.10	0.11	0.11
Longest Drawdown (Days)	316	434	728	728	728

Notes: This table presents the out-of-sample performance of the primary trading strategy using the single best static θ^s of 0.026, determined during the optimization period. Performance is compared against a single-asset (Bitcoin) implementation and three benchmark portfolios: Equally Weighted (EW), Hierarchical Risk Parity (HRP), and Hierarchical Equal Risk Contribution (HERC) (Raffinot 2018). The sample period is from January 2022 to January 2024.

4.2 The Economic Value of an Adaptive Approach

As shown in Table 3, the adaptive meta-learning strategy delivers a Sharpe ratio of 1.34, more than double that of the best-performing static strategy (0.59). This improvement is both statistically and economically significant. To formally test this, we apply the robust bootstrap procedure of Ledoit and Wolf (2008) to the daily returns of both strategies, which yields a p-value of 0.008 for the difference in Sharpe ratios. These findings allow us to reject the null hypothesis (H_0) at the 1% significance level and demonstrate that the predictability of cryptocurrency returns is indeed state-dependent.

The inconsistent monthly performance of the static strategies, shown in Figure 6, visually underscores the need for a dynamic approach, while the cumulative performance in Figure 7 illustrates the economic value generated by the adaptive model over the entire out-of-sample period.



Figure 6: Monthly Returns of Static Base Strategies

Notes: This figure shows the monthly rate of return for the three static base strategies ($\theta^s = 0.01, 0.02, 0.04$) over the out-of-sample period. It highlights that no single parameter is consistently optimal, motivating the need for an adaptive approach.

Table 3: Out-of-Sample Performance: Adaptive vs. Static Strategies

Strategy	RR (%)	MDD (%)	PF	SRPR	SORR	LDD
Panel A: Static Strategies						
Static $\theta^s = 0.01$	63.19	-47.87	1.11	0.59	0.81	261
Static $\theta^s = 0.02$	-30.43	-56.14	1.02	0.11	0.15	458
Static $\theta^s = 0.04$	45.23	-50.50	1.11	0.53	0.73	356
Panel B: Adaptive Meta-Learning Strategy						
Combined Features	445.98	-29.91	1.27	1.34**	1.94	156
Panel C: Benchmark Portfolios						
Equally Weighted (EW)	-19.05	-66.20	1.01	0.07	0.10	728
HRP	-17.79	-65.97	1.02	0.08	0.11	728
HERC	-17.76	-65.95	1.02	0.08	0.11	728

Notes: This table compares the out-of-sample performance of the adaptive meta-learning strategy against the three static base strategies and benchmark portfolios. RR is the total Rate of Return, MDD is the Maximum Drawdown, PF is the Profit Factor, SRPR is the Sharpe Ratio, SORR is the Sortino Ratio, and LDD is the Longest Drawdown Duration in days. The sample period is from January 2022 to January 2024.

** The Sharpe ratio is statistically greater than that of the best-performing static strategy ($\theta^s = 0.01$) at the 1% significance level, based on the bootstrap test of Ledoit and Wolf (2008).

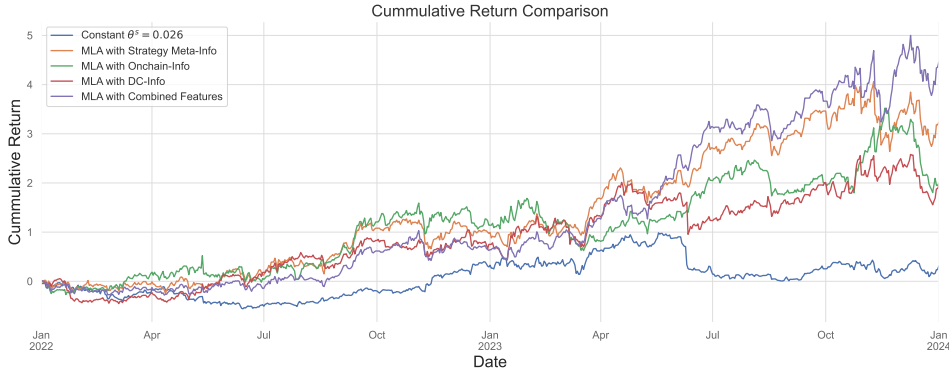


Figure 7: Cumulative Return Comparison: Adaptive vs. Static Strategies

Notes: This figure plots the out-of-sample cumulative log-returns for the adaptive meta-learning strategy (Combined Features) against the best-performing static strategy ($\theta^s = 0.01$) and the meta-learning models trained on individual feature sets.

4.3 Decomposing the Sources of Predictability

Having established the value of an adaptive framework, we now decompose the sources of this predictability to validate our hypothesis that novel information sets are key drivers of inefficiency. We train four separate meta-learning models, each using only one of our proposed feature categories.

The results, presented in Table 4, are twofold. First, traditional sources of information, such as market statistics and strategy performance history, are powerful predictors, with each generating Sharpe ratios above 1.15. Second, and central to this paper’s contribution, our

two novel information sets (On-Chain Information and DC-Derived Indicators) are also highly valuable, generating Sharpe ratios of 0.93 and 0.92, respectively. This confirms that fundamental blockchain activity and event-based trend asymmetries provide unique, economically relevant signals not fully captured by standard financial data.

Crucially, the fact that the combined model (Table 3) outperforms any individual model strongly suggests the feature sets are complementary. As confirmed by a correlation analysis of the models’ predictions (Appendix F), each information set provides a different lens on the market’s state, allowing the integrated model to make more robust and profitable decisions.

Table 4: Performance by Information Set

Meta-Learning Model Feature Set	RR (%)	MDD (%)	PF	SRPR	SORR	LDD
Strategy Meta-Information	323.77	−28.16	1.23	1.17	1.68	109
On-Chain Information	199.04	−39.60	1.18	0.93	1.33	133
DC-Derived Indicators	194.30	−48.24	1.18	0.92	1.30	171
Market Regime Information	372.15	−44.69	1.24	1.19	1.71	323

Notes: This table presents the out-of-sample performance of four separate meta-learning models, each trained using only one of the four specified feature categories. The sample period is from January 2022 to January 2024.

4.4 Robustness Checks

To ensure that our central findings are not an artifact of a specific model specification, we conduct two key robustness checks. First, we test the sensitivity of the final combined model to variations in its key hyperparameters. We find that the superior performance of the adaptive strategy is robust across a range of historical lookback windows ($h_{strategy}$, h_{dc} , etc.) for feature calculation and for different values of the α parameter in our custom weight function. While performance varies, the adaptive model consistently and significantly outperforms the static benchmarks across all tested specifications. Detailed results of this sensitivity analysis are provided in Appendix G.

Second, we verify the economic contribution of the novel weighted loss function introduced in Section 3.4. We re-run the backtests for all meta-learning models using a standard, unweighted loss function. The results confirm the utility of our proposed methodology; incorporating the return-based weights improved the total rate of return in four of the five model specifications. This demonstrates that aligning the model’s training objective more closely with the economic goal of maximizing returns leads to tangible performance gains. A full comparison of the weighted versus unweighted models is available in Appendix H. Finally, to ensure our findings are not specific to the LightGBM algorithm, we replicate the analysis using an alternative XGBoost model. The results, detailed in Appendix I, confirm that the adaptive framework’s outperformance is robust to the choice of machine learning model.

4.5 Economic Interpretation of the Adaptive Model

A critical step beyond establishing statistical performance is to interpret the economic logic learned by the adaptive model. By analyzing which features drive its decisions under different

market conditions, we can verify that the model is capturing genuine, economically intuitive regime shifts rather than merely fitting to noise.

First, we examine the model’s dynamic feature importance over the backtest period, shown in Figure 8. The analysis reveals a clear and economically consistent pattern: the model’s reliance on different information sets evolves in response to market volatility. During the high-volatility environment of early 2022, the model assigned greater importance to DC-Derived Indicators, particularly those capturing trend asymmetries and the magnitude of overshoots. This suggests that in turbulent, uncertain markets, the model prioritizes information about price dynamics and potential reversals. Conversely, as market volatility subsided in late 2022 and 2023, the model’s attention shifted towards fundamental on-chain metrics, such as exchange flows and holder behavior, and strategy-specific meta-information, like recent returns. This behavior is consistent with economic intuition, as fundamental drivers and strategy momentum are likely to be more reliable predictors in more stable, trending market environments. It suggests the model learns to prioritize signals of investor sentiment and panic (captured by trend asymmetries) during crises, while focusing on fundamental value (captured by on-chain metrics) during periods of calm.

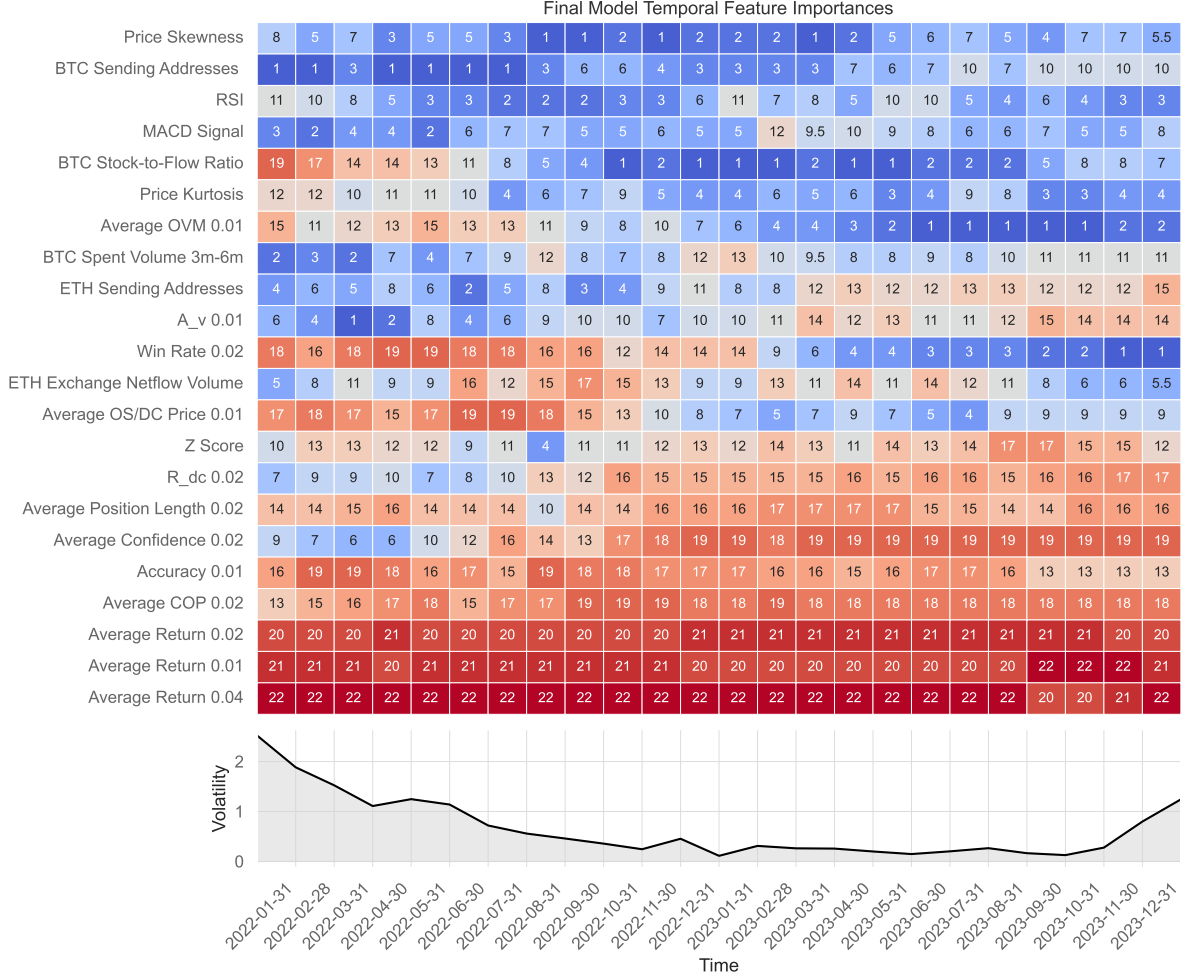


Figure 8: Temporal Feature Importance and Market Volatility

Notes: The top panel displays the time-varying importance ranking of the top features in the combined meta-learning model. The bottom panel plots the portfolio’s realized monthly volatility. The figure illustrates how the model’s feature reliance shifts in response to changing market volatility.

Second, we employ SHAP analysis (Figure 9) to deconstruct the model’s learned decision rules at the feature level. This attribution analysis provides direct insight into the economic logic underpinning its parameter selections. For instance, the model learns to select a smaller θ^s (Class 0), which is better suited for mean-reverting conditions, when it observes a combination of signals indicating an overbought market (a high Z-Score) and potential selling pressure (positive Ethereum exchange netflows). In contrast, it favors a larger θ^s (Class 2), optimal for capturing long trends, when it observes signals of market strength and accumulation, such as extended average position lengths and negative exchange netflows. This learned behavior mirrors principles of behavioral finance, such as reacting to herding signals (exchange flows) and accounting for loss aversion (by adapting the trend sensitivity via θ^s) in different market states. This granular analysis demonstrates that the meta-learning framework is not a “black box”; rather, it has successfully learned to identify and react to interpretable, regime-dependent economic states in a manner consistent with financial theory.

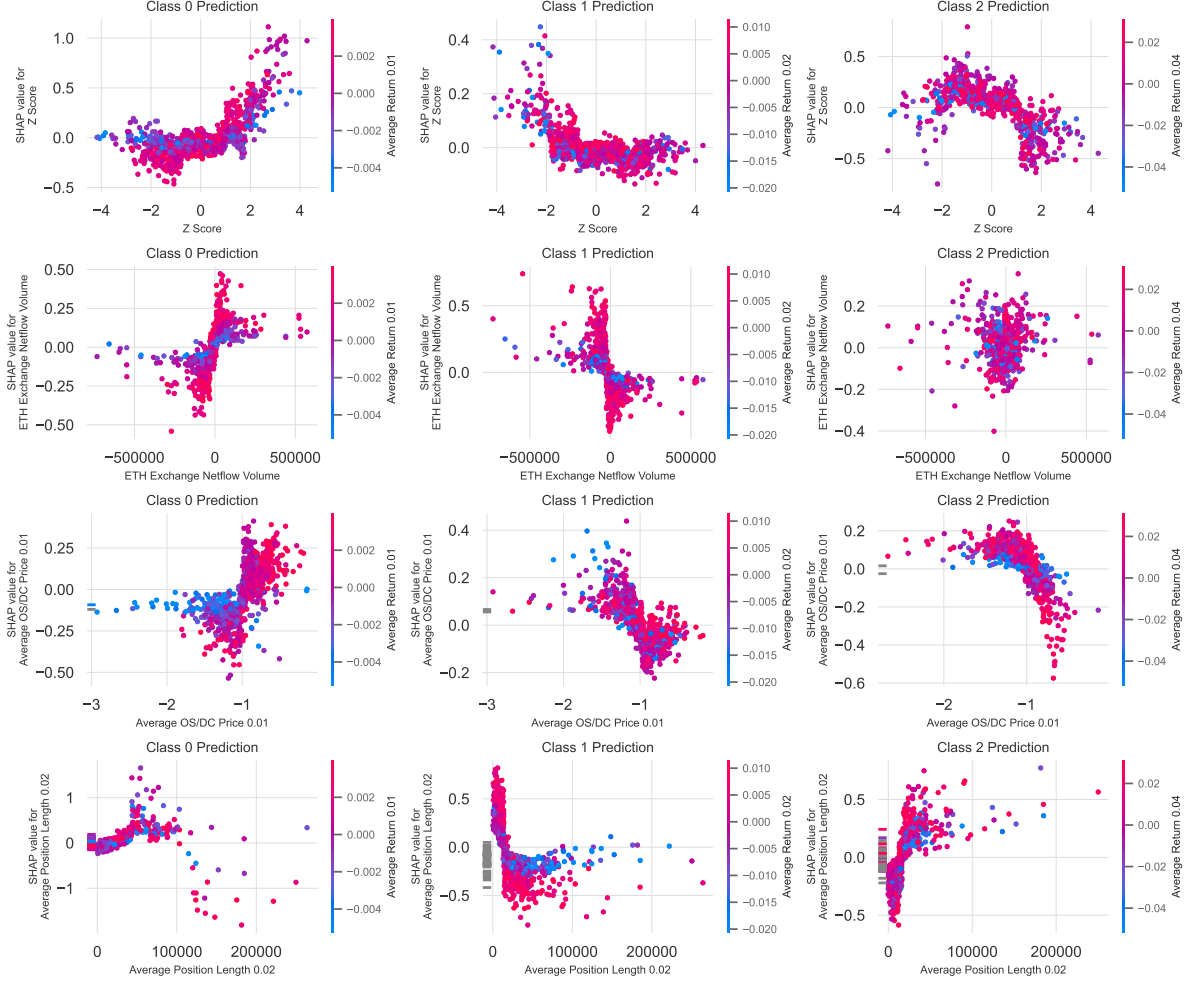


Figure 9: SHAP Analysis of Feature Impact on Class Predictions

Notes: This figure shows SHAP dependence plots for a selection of key features. Each row corresponds to a feature, and each column corresponds to a model output class (i.e., the selected θ^s). The plots reveal how the marginal impact of a feature on the model's decision changes with the feature's value.

5 Discussion

Our findings contribute a potential resolution to the conflicting literature on cryptocurrency market efficiency. Rather than a binary efficient/inefficient state, our results indicate that efficiency is conditional on both the market regime and the information set used for evaluation. The superior performance of our adaptive framework validates the AMH by demonstrating that alpha is a time-varying reward for correctly identifying and exploiting these transient states (Lo 2004; Rönkkö et al. 2024).

5.1 Implications for Market Efficiency Theory

Theoretically, our work supports a nuanced view of efficiency. The predictive power of on-chain data confirms that prices can be slow to incorporate fundamental information about network health, while the significance of DC-derived indicators provides quantitative evidence of behavioral biases manifesting as predictable patterns (Kahneman and Tversky 2013; Mensi

et al. 2025). This suggests that markets can be simultaneously efficient with respect to one information set (e.g., past prices) but inefficient with respect to others (e.g., on-chain flows), with the relevance of each information set changing with the market state.

5.2 Implications for Practitioners

Practically, our findings offer a roadmap for quantitative asset managers. They highlight the value of integrating non-traditional, digitally-native data sources and moving beyond static models. While the reported rates of return are substantial, they reflect the strategy’s high-frequency nature within a leveraged futures market during a particularly volatile period, as detailed in Section 4. For practitioners, the key takeaway is not the absolute return, but the robust evidence that an adaptive approach can more than double risk-adjusted performance while simultaneously reducing downside risk a primary goal of portfolio management.

5.3 Methodological Contributions

This paper also provides a robust methodological toolkit for studying non-stationary markets. The use of SU for clustering offers a superior alternative to correlation for capturing non-linear dependencies. Furthermore, our development of an economically-weighted loss function represents a critical refinement for financial machine learning, ensuring a model’s statistical objective is better aligned with an investor’s economic goals. The fact that a naive volatility-timing strategy fails to replicate our results confirms that our framework’s value comes from its sophisticated, multi-faceted information processing, not from a simple factor exposure.

6 Conclusion

We acknowledge certain limitations that present clear avenues for future research. A primary limitation is that our analysis is presented gross of transaction costs to isolate the strategy’s raw predictive power. Incorporating a realistic model of trading frictions such as exchange fees, funding rates, and price slippage is a complex undertaking. Therefore, a comprehensive study of the strategy’s performance on a net return basis is left for future work due to the scope of this paper.

Methodologically, the framework could be extended in several promising directions. Future work could integrate more advanced statistical regime-detection models, such as Bayesian hidden Markov models, to further refine the adaptive mechanism (Hamilton 1989). Additionally, the use of path signatures from rough path theory offers a powerful, model-free alternative. Path signatures could serve both as a novel feature set to capture complex, path-dependent information missed by traditional indicators, and as a more robust input for the regime-detection model itself. Finally, applying this adaptive framework to other volatile asset classes, such as commodities or emerging market equities, could further validate its generalizability.

This paper provides decisive, out-of-sample evidence that cryptocurrency markets exhibit a dynamic, state-dependent form of efficiency, consistent with the AMH. By developing an adaptive meta-learning framework, we demonstrate that modeling and reacting to market regimes yields more than double the risk-adjusted return of the best-performing static strategy.

Our primary contribution lies in identifying the drivers of this time-varying predictability. We validate two novel, digitally-native information sets fundamental on-chain metrics and behavioral trend asymmetries as economically significant sources of alpha (Mokni et al. 2025). The model's ability to learn intuitive economic rules, such as prioritizing trend signals during market panic and fundamentals during calm, confirms that these inefficiencies are not random noise but structured, interpretable phenomena.

This study provides a definitive framework for investors and quantitative asset managers operating within the rapidly evolving landscape of digital assets. We illustrate that the "pockets of inefficiency" posited by the AMH extend beyond theoretical constructs; rather, they represent structured, interpretable phenomena that can be systematically leveraged for strategic advantage. As market environments evolve and data availability expands, the primary challenge transitions from the age-old debate regarding market efficiency to the identification of innovative informational datasets and adaptive modeling techniques that can effectively navigate these dynamic market conditions. Our findings indicate that the future of alpha generation hinges on addressing this critical question.

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Appendices

Appendix A Detailed On-Chain Features

Table A.1 provides a complete list of the 43 on-chain metrics used as inputs for the meta-learning framework, following the feature selection process described in Section 3.1.2.

Table A.1: List of on-chain features used in this study

Feature	Market	Category
Supply Last Active 3m-6m, Supply Last Active 3y-5y, Supply Last Active 1d-1w	BTC	Supply
Exchange Outflow Volume (Total), Exchange Inflow Volume (Total), Exchange Withdrawals, Change-Adjusted Volume (Median)	BTC	Exchange
Miner Revenue (Block Rewards), Miner Revenue (Total), Hash Rate	BTC	Mining
RHODL Ratio, Realized P/L Ratio, Net Realized Profit/Loss, Hodler Net Position Change, Stock-to-Flow Ratio	BTC	Valuation
Spent Volume 5y-7y, Spent Volume 6m-12m, Spent Volume 1m-3m, Spent Volume 1d-1w, Spent Volume 3m-6m, Spent Outputs 1d-1w	BTC	Spent Outputs and Activity
Sending Addresses	BTC	Transactions
Asia Month-over-Month Price Change, Realized Volatility (3 Months)	BTC	Price Indicators
Supply Last Active 3m-6m, Supply Last Active 1d-1w, Supply Last Active 1w-1m	ETH	Supply
Exchange Withdrawals, Exchange Netflow Volume, Exchange Outflow Volume (Total)	ETH	Exchange
Miner Revenue (Total), Miner Revenue (Fees), Blocks Mined	ETH	Mining
SOPR, MVRV Z Score, Market Cap	ETH	Valuation
Transfer Volume (Mean), Sending Addresses, Transaction Gas Limit (Mean), Gas Price (Mean), Fees (Mean), Gas Used (Median)	ETH	Transactions
EU Month-over-Month Price Change	ETH	Price Indicators

Appendix B Directional Change Algorithm

The procedural logic for identifying DC events, DCC points, and Extreme (EXT) points is formally detailed in Algorithm 1.

Algorithm 1 Directional Change (DC) Event Identification

```
1: Initialization: Set initial trend arbitrarily (e.g., Upward).
2:  $P_{high} \leftarrow P(t_0); P_{low} \leftarrow P(t_0); t_{high} \leftarrow t_0; t_{low} \leftarrow t_0.$ 
3: for each new price  $P_c$  at time  $t$  do
4:   if current trend is Upward then
5:     if  $P_c > P_{high}$  then
6:        $P_{high} \leftarrow P_c; t_{high} \leftarrow t$  ▷ Update extreme high
7:     else if  $P_c \leq P_{high} \times (1 - \theta^s)$  then
8:       Downward DC Event Confirmed at  $(P_c, t).$ 
9:       The point  $(P_{high}, t_{high})$  is retrospectively identified as the EXT point.
10:      current trend  $\leftarrow$  Downward.
11:       $P_{low} \leftarrow P_c; t_{low} \leftarrow t.$ 
12:    end if
13:  else if current trend is Downward then
14:    if  $P_c < P_{low}$  then
15:       $P_{low} \leftarrow P_c; t_{low} \leftarrow t$  ▷ Update extreme low
16:    else if  $P_c \geq P_{low} \times (1 + \theta^s)$  then
17:      Upward DC Event Confirmed at  $(P_c, t).$ 
18:      The point  $(P_{low}, t_{low})$  is retrospectively identified as the EXT point.
19:      current trend  $\leftarrow$  Upward.
20:       $P_{high} \leftarrow P_c; t_{high} \leftarrow t.$ 
21:    end if
22:  end if
23: end for
```

Appendix C Methodological Details for Primary Trading Strategy

C.1 Primary DC Prediction Features

The primary DC prediction model, which forecasts the binary target variable y_i defined in Equation 3, uses four features derived from the DC framework, as established in the literature (Bakhach, Tsang, and Jalalian 2016; Tsang, Tao, and Ma 2015). Their formal definitions are as follows:

- **Total Move Value (TMV):** The price change during a DC trend, normalized by the θ threshold.

$$TMV = \frac{P_{EXT,i}^s - P_{EXT,i-1}^s}{P_{EXT,i-1}^s \times \theta^s} \quad (C.1)$$

- **Time for Completion (T):** The duration of the DC trend, measured in minutes.

$$T = t_{EXT,i}^s - t_{EXT,i-1}^s \quad (C.2)$$

- **Overshoot Value (OSV):** The percentage price change during the Overshoot interval,

normalized by the θ threshold.

$$OSV = \frac{P_c - P_{DCC}^s}{P_{DCC}^s \times \theta^s} \quad (C.3)$$

- **Cross-threshold OS Percentage (COP):** A measure of trend strength that compares the extreme point of a θ^s trend to the confirmation point of a larger θ^b trend.

$$COP = \frac{P_{EXT}^s - P_{DCC}^b}{P_{DCC}^b \times \theta^b} \quad (C.4)$$

Appendix D Distribution of Sample Weights

Figure D.1 provides the empirical motivation for the return-weighted loss function described in Section 3.4.3. The left panel shows the distribution of the daily return difference between the best- and worst-performing static strategies. The long right tail indicates that while performance differences are often small, there are days with very large dispersion, which are economically most significant. The right panel shows the distribution of the resulting sample weights, confirming that the exponential function effectively assigns high importance to these critical observations.

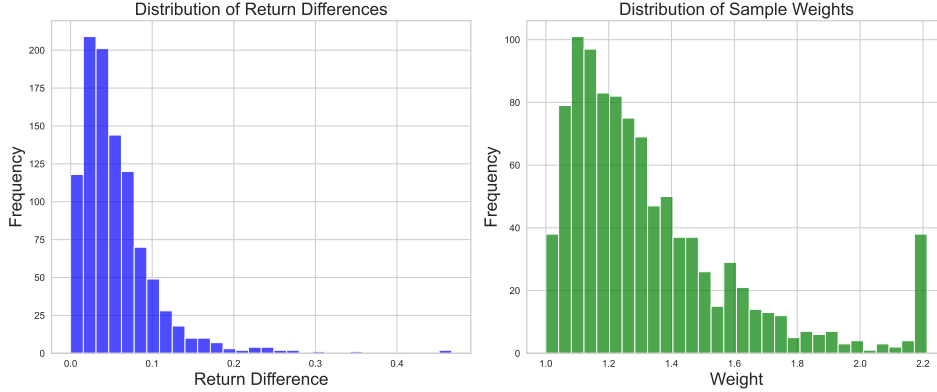


Figure D.1: Histograms of Daily Return Differences and Sample Weights

D.1 Primary Trading Algorithm Logic

Algorithm 2 formally outlines the logic of the two-sided, counter-trend strategy used for all base models.

Algorithm 2 Primary Trading Algorithm Logic

```
1: Assumption: No open positions.  $f$  is the DC prediction model.
2: if No Open Position then
3:   if a  $DC^s$  Event Occurs then
4:      $\hat{y}_i \leftarrow f([TMV, OSV, T, COP])$ 
5:     if  $\hat{y}_i = 0$  then
6:       Open counter-trend position.
7:     else
8:       Wait for next event.
9:     end if
10:  else if a  $DC^b$  Event Occurs and previous  $\hat{y}_i$  was 1 then
11:    Open counter-trend position.
12:  end if
13: else[An open position exists]
14:   if an opposing  $DC^s$  Event Occurs then
15:     Close all open positions.
16:   end if
17: end if
```

Appendix E Performance Evaluation Metrics

This appendix provides the formal definitions for the performance metrics used to evaluate all trading strategies in this study.

- **Rate of Return (RR):** The total percentage gain or loss on the initial capital over the entire evaluation period.

$$RR = \frac{V_T - V_0}{V_0} \quad (\text{E.1})$$

where V_T is the final portfolio value and V_0 is the initial capital.

- **Profit Factor (PF):** The ratio of gross profits to gross losses. A value greater than 1 indicates a profitable strategy.

$$PF = \frac{\sum \text{Profitable Trades}}{|\sum \text{Losing Trades}|} \quad (\text{E.2})$$

- **Maximum Drawdown (MDD):** The largest peak-to-trough decline in portfolio value, representing the maximum unrealized loss.

$$MDD = \max_{t \in [0, T]} \left(\frac{V_{peak}(t) - V(t)}{V_{peak}(t)} \right) \quad (\text{E.3})$$

where $V_{peak}(t)$ is the highest portfolio value recorded up to time t .

- **Longest Drawdown Duration (LDD):** The longest time period, in days, from a portfolio peak to a new peak.

- **Sharpe Ratio (SRPR):** The annualized ratio of the portfolio’s excess return over the risk-free rate to the standard deviation of its returns.

$$SRPR = \frac{\mathbb{E}[R_p - R_f]}{\sigma_p} \quad (\text{E.4})$$

where R_p is the portfolio’s return, R_f is the risk-free rate (assumed to be zero), and σ_p is the standard deviation of the portfolio’s returns.

- **Sortino Ratio (SORR):** Similar to the Sharpe Ratio, but it only penalizes for downside volatility.

$$SORR = \frac{\mathbb{E}[R_p - R_f]}{\sigma_d} \quad (\text{E.5})$$

where σ_d is the standard deviation of negative portfolio returns (downside deviation).

Appendix F Model Output Correlation Analysis

To further validate that the feature sets provide complementary, non-redundant information, we analyze the pairwise correlation of the meta-models’ daily output probabilities. Table F.1 presents the correlation matrix for the predicted probabilities for each of the three classes ($\theta^s = 0.01, 0.02, 0.04$) across the four models trained on individual feature sets. The generally low correlations confirm that the models are learning distinct signals from each information category. This reinforces the value of their integration within the combined model, as each feature set contributes a unique perspective on the prevailing market regime.

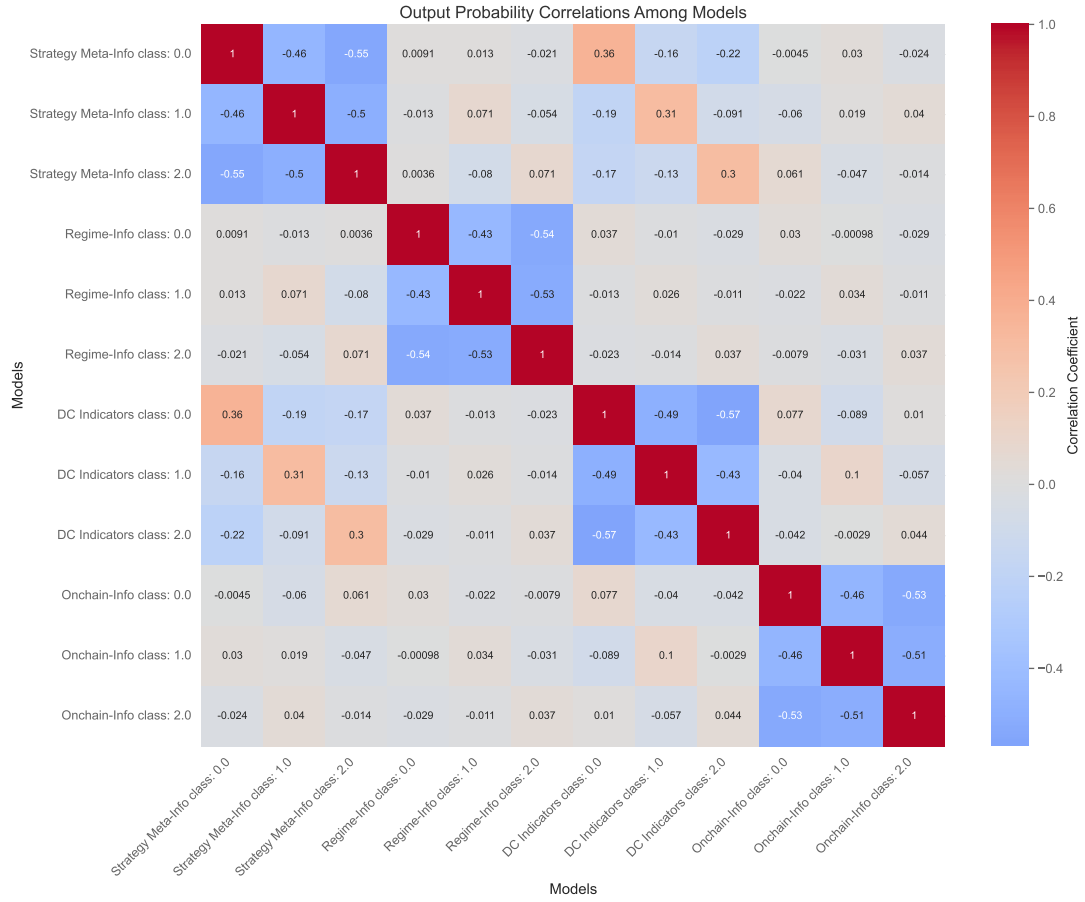


Figure F.1: Correlation Matrix of Meta-Model Output Probabilities

Notes: This figure displays the Pearson correlation matrix of the predicted daily probabilities from the four meta-learning models, each trained on a single feature set. The low correlations indicate that the feature sets provide distinct, complementary predictive signals.

Appendix G Robustness Check: Hyperparameter Sensitivity

Table G.1 presents the results of the sensitivity analysis for the combined meta-learning model's key hyperparameters. The results demonstrate that the strategy's outperformance is robust to variations in the lookback windows for feature engineering and the scaling parameter α in the weighted loss function.

Table G.1: Hyperparameter Sensitivity Analysis

Hyperparameter	Value	RR%	MDD%	PF	SRPR	SORR	LDD
$h_{strategy}$	1*	445.98	−29.91	1.27	1.34	1.94	156
	5	408.68	−60.58	1.25	1.24	1.71	257
	7	225.21	−64.02	1.20	0.97	1.30	253
	10	559.00	−63.46	1.28	1.37	1.91	239
h_{dc}	5	375.91	−37.47	1.26	1.23	1.75	183
	10	247.83	−43.50	1.21	1.03	1.45	184
	20	420.66	−31.32	1.26	1.30	1.92	129
	30*	445.98	−29.91	1.27	1.34	1.94	156
h_{regime}	7	445.88	−29.91	1.27	1.34	1.99	129
	10*	445.98	−29.91	1.27	1.34	1.94	156
	12	371.49	−32.27	1.25	1.22	1.73	237
	15	115.94	−43.92	1.15	0.76	1.03	255
$h_{onchain}$	1*	445.98	−29.91	1.27	1.34	1.94	156
	5	269.70	−31.30	1.21	1.07	1.50	148
	7	336.28	−36.19	1.24	1.19	1.72	143
	10	293.74	−36.14	1.22	1.12	1.60	185
α	5*	445.98	−29.91	1.27	1.34	1.94	156
	10	361.18	−30.70	1.25	1.20	1.72	136
	50	206.70	−31.67	1.19	0.96	1.35	183
	100	313.77	−31.95	1.23	1.14	1.65	111

Notes: This table shows the out-of-sample performance of the combined meta-learning model under different hyperparameter settings. The asterisk (*) denotes the original value used in the main analysis.

Appendix H Robustness Check: Impact of Weighted Loss Function

Table H.1 compares the out-of-sample performance of the meta-learning models trained with and without the economically-weighted loss function. The results show that incorporating the weight function, which prioritizes learning from days with high return dispersion, generally improves the strategy’s profitability.

Table H.1: Performance Comparison With and Without Weighted Loss Function

Strategy	RR%	MDD%	PF	SRPR	SORR	LDD
Panel A: Without Weighted Loss Function						
Strategy Meta-Info	296.85	−33.00	1.22	1.13	1.62	152
On-Chain Info	137.06	−45.25	1.15	0.80	1.12	252
DC Indicators	309.02	−51.95	1.22	1.10	1.54	143
Regime-Info	250.38	−38.52	1.21	1.04	1.42	182
Combined Features	496.57	−37.54	1.28	1.37	2.02	143
Panel B: With Weighted Loss Function						
Strategy Meta-Info	323.77	−28.16	1.23	1.17	1.68	109
On-Chain Info	199.04	−39.60	1.18	0.93	1.33	133
DC Indicators	194.30	−48.24	1.18	0.92	1.30	171
Regime-Info	372.15	−44.69	1.24	1.19	1.71	323
Combined Features	445.98	−29.91	1.27	1.34	1.94	156

Notes: This table compares the performance of the meta-learning models when trained with a standard (unweighted) loss function versus the economically-weighted loss function proposed in this paper.

Appendix I Robustness Check: Alternative Machine Learning Model

To ensure our findings are not contingent on the specific choice of the LightGBM algorithm, we replicate our main analysis using an alternative, well-established model: Extreme Gradient Boosting (XGBoost). Table I.1 compares the out-of-sample performance of the best static strategy against an adaptive meta-learning framework where the LightGBM model is replaced with a tuned XGBoost model. The results confirm that our central finding is robust to the choice of machine learning algorithm. The adaptive XGBoost model delivers a Sharpe ratio of 1.28, substantially outperforming the static benchmark and demonstrating a significant, economically meaningful improvement in risk-adjusted returns. While the LightGBM model shows slightly stronger performance and was thus retained for the main analysis, the robust outperformance of the XGBoost model validates that the economic value stems from the adaptive framework itself, rather than an idiosyncrasy of a particular algorithm.

Table I.1: Performance Comparison with Alternative (XGBoost) Model

Strategy	RR%	MDD%	PF	SRPR	SORR	LDD
Panel A: Static Strategy						
Best Static ($\theta^s = 0.01$)	63.19	−47.87	1.11	0.59	0.81	261
Panel B: Adaptive Meta-Learning Strategy						
Adaptive XGBoost Model	412.33	−31.54	1.26	1.28	1.85	168

Notes: This table presents the out-of-sample performance of an adaptive meta-learning strategy using an XGBoost model, compared against the best-performing static strategy. The sample period is from January 2022 to January 2024.

Appendix J Meta-Learning Feature Definitions

This appendix provides the complete mathematical definitions for the features used in the meta-learning model, as described in Section 3.4.2. Features are calculated over a historical window of h days.

J.1 Strategy Meta-Information Features

These metrics are calculated for each base strategy (i.e., for each $\theta^s \in \Theta$).

- **Number of Orders** (N_{orders}): The total count of positions taken in the past h days.
- **Average Position Length**: The average duration, in minutes, for which a position remains open.

$$\text{Average Position Length} = \frac{1}{N_{orders}} \sum_{i=1}^{N_{orders}} (t_{exit,i} - t_{enter,i}) \quad (\text{J.1})$$

where $t_{enter,i}$ and $t_{exit,i}$ are the entry and exit times for order i .

- **Long Ratio**: The ratio of long orders to the total number of orders.

$$\text{Long Ratio} = \frac{1}{N_{orders}} \sum_{i=1}^{N_{orders}} \mathbb{I}(\text{OrderType}(i) = \text{Long}) \quad (\text{J.2})$$

- **Win Rate**: The ratio of profitable orders to the total number of orders.

$$\text{Win Rate} = \frac{1}{N_{orders}} \sum_{i=1}^{N_{orders}} \mathbb{I}(r_i > 0) \quad (\text{J.3})$$

where r_i is the return for order i .

- **Average Return**: The mean return across all orders.

$$\text{Average Return} = \frac{1}{N_{orders}} \sum_{i=1}^{N_{orders}} r_i \quad (\text{J.4})$$

- **DC Type Ratio**: The ratio of orders triggered by a large trend confirmation (DC^b) to the total number of orders.

$$\text{DC Type Ratio} = \frac{1}{N_{orders}} \sum_{i=1}^{N_{orders}} \mathbb{I}(\text{EntryCondition}(i) = DC^b) \quad (\text{J.5})$$

- **Primary Model's Accuracy**: The accuracy of the base DC prediction model.

$$\text{Accuracy} = \frac{1}{N_{DC^s}} \sum_{i=1}^{N_{orders}} \mathbb{I}(y_i = \hat{y}_i) \quad (\text{J.6})$$

where N_{DC^s} is the number of small DC events, and y_i and $\hat{y}_i$ are the true and predicted labels.

- **Average Confidence:** The average prediction probability from the base DC prediction model.

$$\text{Average Confidence} = \frac{1}{N_{orders}} \sum_{i=1}^{N_{orders}} p_{i,prob} \quad (\text{J.7})$$

where $p_{i,prob}$ is the output probability for sample i .

J.2 DC-Derived Indicators

These indicators provide a deeper view of market dynamics using the DC framework and are calculated for each base strategy (i.e., for each $\theta^s \in \Theta$).

- **Number of DC events (N_{DC}):** Total number of DC events during the past h days (Aloud 2016).
- **Average OS Duration (OVD):** Average duration of the Overshoot intervals (Aloud 2016).

$$\text{Average OVD} = \frac{1}{N_{DC}} \sum_{i=2}^{N_{DC}} (t_{EXT,i}^s - t_{DCC,i-1}^s) \quad (\text{J.8})$$

- **Average OS Magnitude (OVM):** Average price change during the Overshoot intervals (Aloud 2016).

$$\text{Average OVM} = \frac{1}{N_{DC}} \sum_{i=2}^{N_{DC}} |P_{EXT,i}^s - P_{DCC,i-1}^s| \quad (\text{J.9})$$

- **Average Trend Duration:** Average duration of the DC intervals (from an extreme point to its confirmation) (Aloud 2016).

$$\text{Average Trend Duration} = \frac{1}{N_{DC}} \sum_{i=1}^{N_{DC}} T_i \quad (\text{J.10})$$

- **Coastline:** The sum of all normalized trend magnitudes (TMV) during the past h days (Aloud et al. 2012).

$$\text{Coastline} = \sum_{i=1}^{N_{DC}} |TMV_i| \quad (\text{J.11})$$

- **Average Magnitude:** The average absolute price change between consecutive DC confirmation points (Aloud 2016).

$$\text{Average Magnitude} = \frac{1}{N_{DC}} \sum_{i=2}^{N_{DC}} |P_{DCC,i}^s - P_{DCC,i-1}^s| \quad (\text{J.12})$$

- **Average Time Adjusted Return of DC (R_{DC}):** The ratio of price change to duration

for a DC trend (Tsang, Tao, and Ma 2015).

$$R_{DC} = \frac{1}{N_{DC}} \sum_{i=1}^{N_{DC}} \frac{|TMV_i| \times \theta^s}{T_i} \quad (J.13)$$

- **Average OS to DC Price Ratio:** The average ratio of the price change during an Overshoot to the price change during the preceding DC interval (Kampouridis and Otero 2017).

$$\text{Average OS/DC Price Ratio} = \frac{1}{N_{DC}} \sum_{i=2}^{N_{DC}} \frac{P_{EXT,i}^s - P_{DCC,i-1}^s}{P_{DCC,i-1}^s - P_{EXT,i-1}^s} \quad (J.14)$$

- **Average OS to DC Duration Ratio:** The average ratio of the duration of an Overshoot to the duration of the preceding DC interval (Kampouridis and Otero 2017).

$$\text{Average OS/DC Time Ratio} = \frac{1}{N_{DC}} \sum_{i=2}^{N_{DC}} \frac{t_{EXT,i}^s - t_{DCC,i-1}^s}{t_{DCC,i}^s - t_{EXT,i}^s} \quad (J.15)$$

- **Uptrend and Downtrend Asymmetry in Time (A_T):** The relative difference in the median duration of uptrends versus downtrends (Tao 2018).

$$A_T = \frac{T_{median,\uparrow} - T_{median,\downarrow}}{T_{median,\uparrow} + T_{median,\downarrow}} \quad (J.16)$$

- **Uptrend and Downtrend Asymmetry in Returns (A_R):** The relative difference in the median time-adjusted return (R_{DC}) of uptrends versus downtrends (Tao 2018).

$$A_R = \frac{R_{DC,median,\uparrow} - R_{DC,median,\downarrow}}{R_{DC,median,\uparrow} + R_{DC,median,\downarrow}} \quad (J.17)$$

- **Average COP:** The average value of the Cross-threshold OS Percentage feature over the past h days.

$$\text{Average COP} = \frac{1}{N_{DC}} \sum_{i=1}^{N_{DC}} COP_i \quad (J.18)$$

- **Uptrend and Downtrend Volume Asymmetry (A_v):** This indicator captures the asymmetry of traded volume in uptrends versus downtrends. First, we define "Volume per Trade" (vt) as the average volume per trade during a single DC trend:

$$vt = \frac{1}{N_{trades}} \sum_{j=1}^{N_{trades}} V_j \quad (J.19)$$

where V_j is the volume of trade j , and N_{trades} is the total number of trades that occurred during the DC trend interval (between $t_{EXT,i-1}^s$ and $t_{EXT,i}^s$). The volume asymmetry indicator A_v is then defined as:

$$A_v = \frac{vt_{median,\uparrow} - vt_{median,\downarrow}}{vt_{median,\uparrow} + vt_{median,\downarrow}} \quad (J.20)$$

where $vt_{median,\uparrow}$ and $vt_{median,\downarrow}$ represent the median volume per trade in uptrends and downtrends, respectively, over the historical window.

J.3 Market Regime Information Features

These statistical metrics are calculated on the aggregated portfolio price series over a historical window of h days.

- **Relative Strength Index (RSI):** A momentum oscillator that measures the speed and change of price movements to identify overbought or oversold conditions (Chong, Ng, and Liew 2014). It is calculated as follows over a 14-period lookback:

$$\text{Average Gain} = \frac{1}{14} \sum_{i=1}^{14} \max(\Delta P_i, 0) \quad (\text{J.21})$$

$$\text{Average Loss} = \frac{1}{14} \sum_{i=1}^{14} \max(-\Delta P_i, 0) \quad (\text{J.22})$$

$$RS = \frac{\text{Average Gain}}{\text{Average Loss}} \quad (\text{J.23})$$

$$RSI = 100 - \frac{100}{1 + RS} \quad (\text{J.24})$$

where ΔP_i is the price change. We use the average RSI over the past h days.

- **Moving Average Convergence/Divergence (MACD) Signal:** A trend-following momentum indicator (Chong, Ng, and Liew 2014).

$$\text{MACD Line} = \text{EMA}_{12}(\text{Price}) - \text{EMA}_{26}(\text{Price}) \quad (\text{J.25})$$

$$\text{MACD Signal Line} = \text{EMA}_9(\text{MACD Line}) \quad (\text{J.26})$$

where EMA is the Exponential Moving Average. We use the average of the MACD Signal Line over the past h days.

- **Price Standard Deviation (Volatility):** The standard deviation of prices over the window.

$$\sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (P_i - \mu)^2} \quad (\text{J.27})$$

where μ is the average price and n is the number of observations in the window.

- **Price Skewness:** The asymmetry of the price distribution.

$$\text{Skewness} = \frac{\frac{1}{n} \sum_{i=1}^n (P_i - \mu)^3}{\sigma^3} \quad (\text{J.28})$$

- **Price Kurtosis:** The "tailedness" of the price distribution.

$$\text{Kurtosis} = \frac{\frac{1}{n} \sum_{i=1}^n (P_i - \mu)^4}{\sigma^4} \quad (\text{J.29})$$

- **Z-Score:** The number of standard deviations the current price is from the historical mean.

$$Z = \frac{P_{current} - \mu}{\sigma} \quad (\text{J.30})$$